

Coupling Proof :

The octopus arm currently contains 6 links with one as fixed and the movement for hold/grip is done by tendon and pulley mechanism. The tendon is passed through holes of each part and when tendon is pulled, due to tension the octopus arms grip

- Assumptions :

tendon is frictionless , tension is uniform along the length, spool converts rotation to tendon pulling.

- Proof:

$$M = T r_i = k(\theta_i)$$

$r_i \Rightarrow$ distance of hole from joint axis.
 $R =$ spool radius, $\theta_i \Rightarrow$ angle movement of each part, $T \Rightarrow$ tension, $k \Rightarrow$ rotational stiffness, $\Delta l =$ pulled length

$$\therefore \theta_i = \frac{T \cdot r_i}{k} \quad ; \quad \Delta l_{\text{total}} = \sum r_i \theta_i = \frac{T}{k} \sum (r_i)^2$$

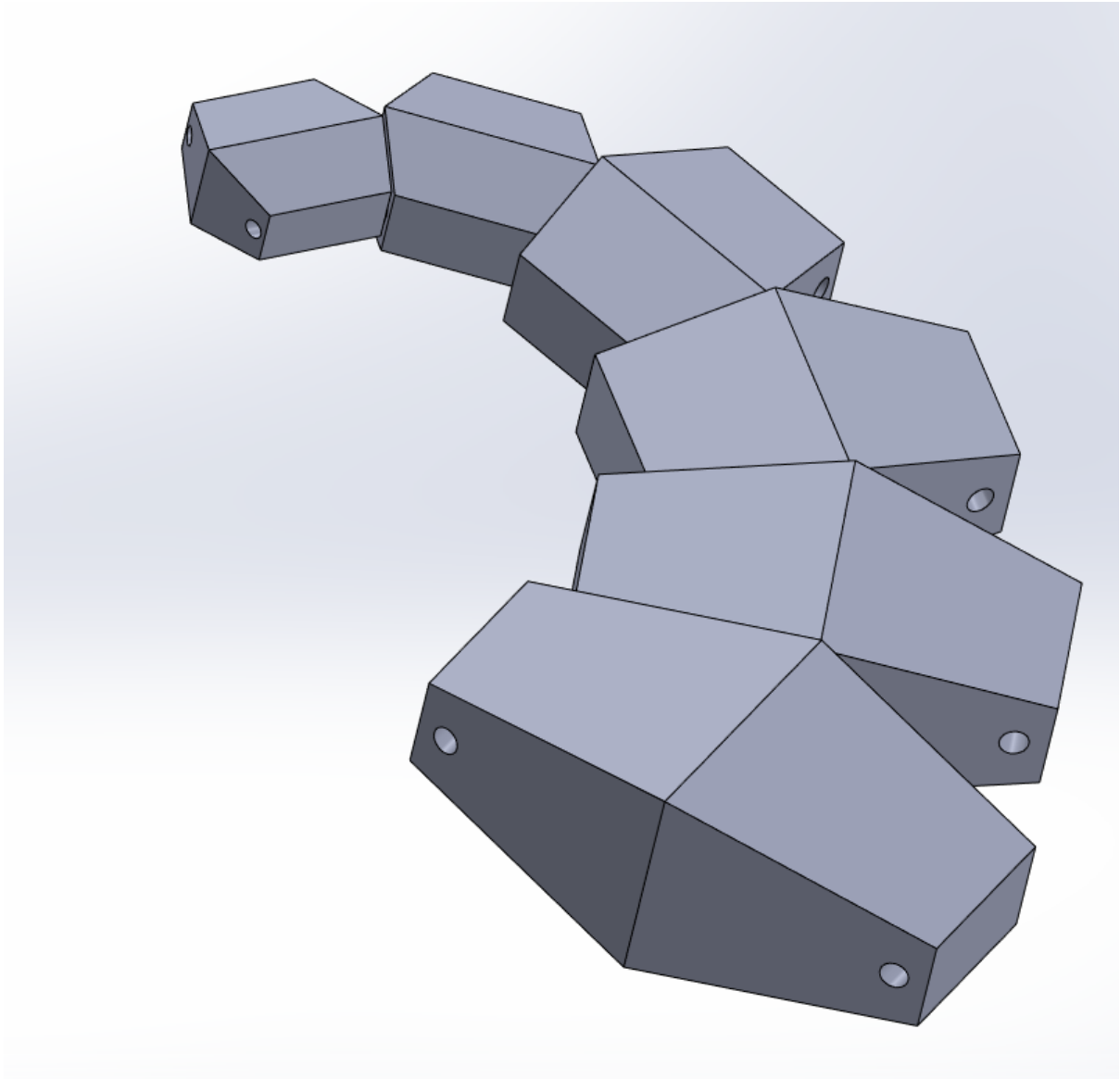
Also, $\Delta l = R \phi$

$$\therefore \theta_i = \text{clamp} \left[\left(\frac{R \cdot r_i}{\sum r_i^2} \right) \cdot \phi, 0, \theta_{\max} \right]$$

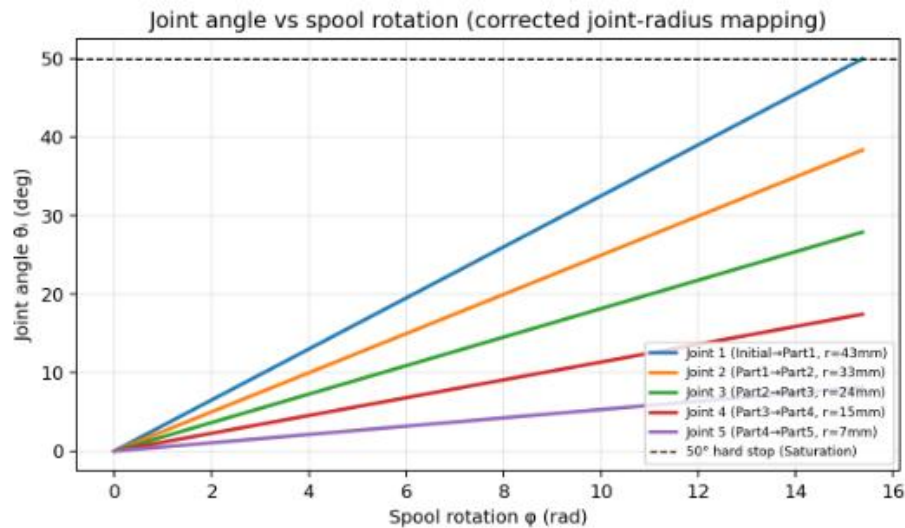
Joint	r_i	Gain = $R \cdot r_i / \sum r_i^2$	$\phi_{50\% \text{ stop}}$
Initial part $\rightarrow$ Part 1	43	0.05676	15.38
Part 1 $\rightarrow$ Part 2	33	0.04356	20.03
Part 2 $\rightarrow$ Part 3	24	0.03168	27.55
Part 3 $\rightarrow$ Part 4	15	0.01980	44.07
Part 4 $\rightarrow$ Part 5	7	0.00924	94.45

Therefore to maintain mechanical sanity (no part intervention with other, spool angle rotation is max taken to be 15.38deg.

Curl Sequence Plot :



This is the position view when all the parts are rotated to their maximum limit and for smaller object and better grip can add more parts to elongate the arm and have complete surface hold/grip of object.



Adapting the design :

The current exact design is not suited to smaller / delicate objects as when the arm will grip around the object it will not be able to hold it and even damage it.

But, Larger object can be gripped inside the current arm but it should not be even more larger than the grip surrounding capacity.

So like if to be modified this design according to adaptability situations, would need more components to join as tail and redesign shape for holding delicate objects.